THE UNIVERSITY OF CHICAGO

Master of Science in Applied Data Science

Course Syllabus

COURSE TITLE: Causal Models in Data Science

ADSP Course Code: #32029

TERM Year; Online: Autumn 2025 Online

Synchronous Class: On Wednesdays from 6 PM to 9 PM Central Time

Instructor Name: Jeong-Yoon Lee

Instructor Email Address: jeongyoon@uchicago.com

TA Name: Akshat Palnitkar

TA Email Address: akshatpalnitkar@uchicago.edu

Responsiveness: When contacting your instructor or TA with questions, please allow 24-48 hours for a response. Technical questions about assignments and coursework should be addressed to the TA first, with the instructor cc'd on the message if necessary. You are encouraged to attend all TA sessions and office hours to get your questions answered.

COURSE DESCRIPTION

This 10-week course introduces students to the field of causal machine learning, focusing on the distinction between correlation and causation. The course emphasizes applying causal inference techniques using Python. It covers both foundational concepts and applied methods for estimating causal effects in various real-world domains such as healthcare, economics, and marketing.

Each class will feature a lecture, an invited professional speaker, and a lab session.

PREREQUISITES

To be successful in the course, the following are the prerequisites.

- ADSP 31017 - Machine Learning I

COURSE MATERIALS

Required:
- [CIP] Causal Inference in Python by Matheus Facure

Suggested:
- [CIBT] Causal Inference for the Brave and True by Matheus Facure
- [CML] Causal ML Book by Victor Chernozhukov et al.
- [CISSBS] Causal Inference for Statistics, Social, and Biomedical Sciences by Guido Imbens and Donald Rubin
- [MHE] Mostly Harmless Econometrics by Joshua Angrist and Jörn-Steffen Pischke
- [FCCI] A First Course in Causal Inference by Peng Ding
- [MLCI] Machine Learning & Causal Inference: A Short Course by Susan Athey et al.

SOFTWARE & TOOLS
- Python

CLOUD COMPUTING SERVICES
Not applicable.

LEARNING OUTCOMES
By the end of this course, students will:
1. Understand the principles of causal inference and how it differs from correlation-based approaches.
2. Apply causal discovery techniques to infer causal relationships from data.
3. Implement methods such as propensity score matching, instrumental variables, and causal forests to estimate causal effects using Python.
4. Address confounding, mediation, and selection bias in real-world data.
5. Gain practical experience through case studies and projects.

METHODS OF INSTRUCTION

[Autumn 2025, Online via Zoom, Wednesdays, 6 PM - 9 PM CT]

This is a 10-week course with asynchronous and synchronous activities. We will meet weekly for online synchronous class sessions, and I will provide online office hours. Synchronous online sessions will include discussion and instructional activities. Students are expected to review the week's lectures ***prior*** to each synchronous session in order to fully participate in class. Weekly attendance and participation are expected. In addition, students should plan to log into Canvas at least twice a week.

Asynchronous work includes a review of course content, completion of weekly assignments, and collaboration with classmates. Asynchronous content may include video lectures, readings, supplemental videos, discussions, quizzes, homework, and completion of a final project.

TA sessions will be arranged as a group and held on a weekly basis. The TA will be available to provide additional instruction on the course material and provide support for homework.

There are no University breaks during the ten weeks, so it's important not to fall behind with the coursework.

EVALUATION

1. Weekly Assignments (40%): Quiz (15%), Lab (15%), Assignment (10%)
2. Final Project (50%): Proposal (10%), Report (10%), Presentation (30%)
3. Participation (10%)

FINAL PROJECT

Students will plan and conduct a full causal analysis using public datasets, employing multiple techniques covered in the course. Students will be required to submit the project proposal by Week 6, submit the final report by Week 10, and present the project at the last class meeting.

GRADING SCALE

You must earn a grade of B- or better in each of the courses that apply toward your degree requirement.

A =	93%–100%
A- =	90%–92%
B+ =	87%–89%
B =	83%–86%
B- =	80%–82%
C+ =	77%–79%
C =	73%–76%
C- =	70%–72%
F =	0%–69%

COURSE SCHEDULE

Academic quarters consist of 9 weeks of instruction, with the 10th week for assessment or course rescheduling. Refer to the university's academic calendar for quarterly start and end dates.

Important Note: Changes may occur to the syllabus at the instructor's discretion. When changes are made, students will be notified via email and in-class announcement.

Week	Date	Lecture	Invited Talk	Assignments Due
Week 1	10/1	Intro: Causal Inference and Machine Learning (slides)	Causal Inference in Banking: Mixing Applied Micro with Machine Learning by Matheus Facure, Nubank	[CIP] Preface, Ch 1 [MLCI] Video 1 Quiz 1
Week 2	10/8	ATE Estimation: Randomized Experiments, Stats Review (slides)	Causal Econometrics in Tech by Apoorva Lal, Amazon	[CIP] Ch 2 Quiz 2, Lab 1, Assignment 1
Week 3	10/15	ATE Estimation: Linear Regression, Propensity Score, Doubly Robust Estimator (slides)	Applications of Causal Machine Learning in Building a Unified Metric System by Wenjing Zheng, Roblox	[CIP] Ch 4, 5 Quiz 3, Lab 2, Assignment 2
Week 4	10/22	CATE Estimation: Transformed Outcome, Meta-learners, Double ML (slides)	Estimating Network Effects: Some case studies in marketing and product analytics by Huigang Chen, Google	[CIP] Ch 6, 7 Quiz 4, Lab 3-1/3-2, Assignment 3
Week 5	10/29	Causal Identification: Graphical Causal Models (slides)	Ad Measurement & Causal Machine Learning at Snap by Jing Pan, Snap	[CIP] Ch 3 Quiz 5, Lab 4-1/4-2, Assignment 4
Week 6	11/5	LATE Estimation: Instrumental Variable, 2SLS (slides)	Causal Inference in Multi-Sided Marketplaces by Sourav Sinha, Instacart	[CIP] Ch 11 Quiz 6, Lab 5 Final Project Proposal,
Week 7	11/12	Causal Inference with Panel Data: Diff-in-Diff (slides)	Uplift Modeling in Marketing by Jeong-Yoon Lee, Uber	[CIP] Ch 8 Quiz 7, Lab 6, Assignment 5
Week 8	11/19	Causal Inference with Panel Data: Synthetic Control (slides)	Three Fallacies of Applied Causal Inference With Reflections on the Socratic Approach by Totte Harinen, Airbnb	[CIP] Ch 9, 10a Quiz 8, Lab 7, Assignment 6
Week 9	12/3	Switchback. Next steps (slides)	Applied Causal Inference at Netflix by Kevin Mercurio, Netflix	[CIP] Ch 10b, 12 Quiz 9, Lab 8,

				Assignment 7
Week 10	12/10	Final Project Presentation		Lab 9 Final Project Report

ATTENDANCE

This course will meet once a week. Your attendance is required and paramount to your success in this class. You are allowed to miss at most two sessions, provided that you make arrangements with the instructor in advance.

LATE WORK

All assignments must be submitted to this course's Canvas site on the due dates as per the course calendar. If you turn in an assignment late, 10% credit will be deducted from the total score for each day after the deadline. Assignments turned in more than one week late will not receive credit. In the case of unexpected events, you must contact me before the assignment due date in order to receive a grace period. Students can only receive up to 1 grace period in the course.

REQUESTING REASONABLE ACCOMMODATIONS

The University of Chicago is committed to ensuring equitable access to our academic programs and services. Students with disabilities who have been approved for the use of academic accommodations by Student Disability Services (SDS) and need a reasonable accommodation(s) to participate fully in this course should follow the procedures established by SDS for using accommodations. Timely notifications are required in order to ensure that your accommodations can be implemented. Please meet with me to discuss your access needs in this class after you have completed the SDS procedures for requesting accommodations.

Phone: (773) 702-6000
Email: disabilities@uchicago.edu

Please follow the accommodation implementation instructions provided by the disability liaison in the division after you have completed the SDS procedures for requesting accommodations.

You may want to begin by reading through the information published on the Student Disability Services website.

ACADEMIC HONESTY & PLAGIARISM

It is contrary to justice, academic integrity, and to the spirit of intellectual inquiry to submit another's statements or ideas of work as one's own. To do so is plagiarism or cheating, offenses punishable under the University's disciplinary system. Because these offenses undercut the distinctive moral and intellectual character of the University, we take them very seriously.

Proper acknowledgment of another's ideas, whether by direct quotation or paraphrase, is expected. In particular, if any written or electronic source is consulted and material is used from that source, directly

or indirectly, the source should be identified by author, title, and page number, or by website and date accessed. Any doubts about what constitutes "use" should be addressed to the instructor.

Review the University of Chicago Student Manual Academic Honesty and Plagiarism policy.
Review the University of Chicago International Affairs Plagiarism website.
Review the University of Chicago Libraries site on copyright.

ACADEMIC HONESTY & GENERATIVE AI

All assignments in this class may be completed with the aid of AI tools or other online reference sources. Within each submitted assignment, list your assisting sources (by product name and version if an AI tool). Failure to disclose your assisting sources may result in a penalty or a zero grade for the assignment. However, each student's work must remain their own. Discussion of the assignment with classmates in a study group or during office hours is permitted, but your assignment submissions should be unique and should not appear to be substantially copied from other students. Failure to work independently may result in a penalty or a zero grade for the assignment.

EXPECTATIONS FOR STUDENTS

As a student, you will get as much or as little out of this course as the effort you put in. My expectation is that you become involved in the class through participation in Discussions, synchronous class sessions, email, etc. and show respect to all participants in the class. Time management is important to stay focused on assignments and complete them by the required due dates. Please communicate any concerns or issues to the instructor in order to gain a better understanding of the content and stay focused in the class.

Virtual classroom conduct

When you attend class, make sure that you are in a quiet, preferably private space where you can focus and learn productively. It is expected that students will have their cameras on for the duration of the live class session via Zoom. The purpose of this is to promote engagement and help you connect more readily with your instructor, TA, and peers.

Class recordings

Synchronous class sessions will be recorded in Zoom and stored in the course Canvas site for 90 days.

Academic Integrity

Sharing course content is strictly prohibited. Under no circumstances are students permitted to:

- Download asynchronous lecture videos;
- Share sample code on open-source websites;
- Plagiarize individual assignments;
- Discuss course content, whether verbally, or electronically, with future cohorts of Online MS in Applied Data Science students.

INSTRUCTOR COMMITMENTS

I will provide guidance on the course materials and return graded work with constructive feedback in a timely manner, within one week from the assignment due date. Assignments submitted late will receive feedback within 72 hours of their submission. I will make every effort to respond to your questions within 24 hours. I will be available to meet with you individually by appointment. As a part-time instructor, I have a day job. I request you contact me via email for most situations or questions..

COURSE EVALUATION

Towards the end of the course, you will receive an email from the Office of the University Registrar reminding you to provide feedback on the course. You will receive consistent reminders throughout the period when the evaluation is open, and the reminders will stop once you have completed the evaluation. Please note:

- The evaluation is completely anonymous. When the results are released, instructors and departments will not be able to tell which student provided the individual feedback.
- Because it is anonymous and the results are not released to faculty or departments until after grades have been submitted, the feedback will not impact a student's grade.
- The feedback is important so that the instructor can gain insight in to how to improve their teaching and the department can learn how best to shape the curriculum.

STATEMENT OF INTENT

By remaining in this course, you are agreeing to accept this syllabus as a contract and to abide by the guidelines outlined in the document. You will be notified should there be a necessary change to the syllabus.

ONLINE PROGRAM TECHNOLOGY REQUIREMENTS

Set Up and Equipment

For this program, we will be using the video-conferencing platform Zoom. When you join a meeting in Zoom for the first time, you will need to install the program. In order to access the Zoom synchronous sessions, you are required to log into Zoom with your CNET.

If you have not already downloaded Zoom, visit the Download Center to download the Zoom client before the program begins. We encourage you to upgrade to the latest available version of Zoom for new features and bug fixes. You can update your Zoom Desktop Client online here.

Preparation for Joining the Zoom Sessions

We recommend you test Zoom before joining the program's first session so that you get a chance to see the interface and test your webcam and microphone. Please click on this link to download Zoom and join a test meeting.

How do I test my audio?
How do I test my video?

If you're having technical problems with Zoom, 24/7 support is available: call 888-799-9666 ext 2. For other additional support options, please visit the Zoom Help Center or consult these Zoom Technical Support Guidelines.

Guidelines for Recording Students During Online Classes

In this class software will be used to record live class discussions. As a student in this class, your participation in live class discussions will be recorded. These recordings will be made available only to students enrolled in the class, to assist those who cannot attend the live session or to serve as a resource for those who would like to review content that was presented. All recordings will become unavailable to students in the class 90 days after the recording date.